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SOFTWARE ENGINEERING

M02 S03: Requirements Engineering

Today’s session

M02
SESSION 3

01 Requirements and their sources

02 Business, functional and quality requirements

03 Clear wording and acceptance criteria

04 A small requirements register

05 QuickBite M02 Task 2

25–30 minutes of theory and examples, followed by team project work

Project vision and requirements

M01: the project direction

Problem and intended improvement
Stakeholders and their needs
Scope and business objectives

M02: explicit expectations

Required system behaviour
Required qualities
Evidence of acceptance

What is a requirement?

Quick classification exercise - discuss in groups

State, which describes:

- **business goal**
- **system behaviour**
- **quality expectation**
- **constraint**

A Reduce the time staff spend handling reservations.

B 95% of searches finish within 2 seconds under the stated load.

C The existing payment provider must remain.

D Customers can cancel a reservation before pickup.

Which type is each statement? Explain your reason.

Quick classification exercise - discuss in groups

State, which describes:

- **business goal**
- **system behaviour**
- **quality expectation**
- **constraint**

A - business goal	Reduce the time staff spend handling reservations. - **The intended business improvement. Add a measure and target or direction.**
B - quality expectation	95% of searches finish within 2 seconds under the stated load. - **Performance under defined measurement conditions.**
C - constraint	The existing payment provider must remain. - **an imposed limit on solution choices.**
D - system behaviour	Customers can cancel a reservation before pickup. - **A system capability. Clarify when cancellation is permitted.**

Which type is each statement? Explain your reason.

What is a requirement?

A requirement states a needed capability, quality or constraint.

Stakeholder need: “I need reliable vehicle availability.”

System requirement: “The system shall exclude vehicles marked under maintenance from availability search results.”

Types of requirements

Business objective	Functional requirement	Quality requirement / NFR
What improvement matters?	What must the system do?	How well must it perform?
Reduce reservation-handling time.	**Search vehicles by branch and rental period.**	**Return 95% of searches within 2 seconds under stated load.**

Numerical targets are illustrative assumptions.

Types of requirements - constraint as a complementary condition

Business objective

What improvement matters?

Reduce reservation-handling time.

Functional requirement

What must the system do?

Search vehicles by branch and rental period.

Quality requirement / NFR

How well must it perform?

Return 95% of searches within 2 seconds under stated load.

Constraint

What are the limitations?

The existing payment provider must remain.

Numerical targets are illustrative assumptions.

Functional requirements - What must the system do?

FR-01	**The system shall search available vehicles for a selected branch and rental period.**
FR-02	**The system shall reject a reservation that overlaps an existing confirmed reservation for the same vehicle.**
FR-03	**The system shall allow authorised branch staff to mark a vehicle as under maintenance.**

Quality requirements / Non-Functional Requirements - How well must it perform?

A quality requirement needs a metric, conditions and a target.

Weak wording:

The search should be fast.

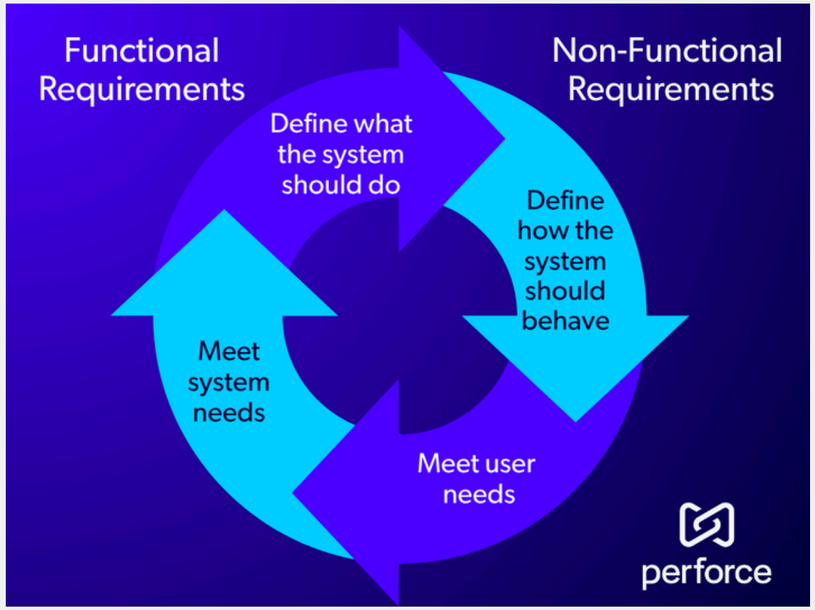
Clearer wording:

At least 95% of availability searches shall finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.

Metric + conditions + target. Numerical values are illustrative assumptions.

Quality requirements / Non-Functional Requirements (NFR)

Consist of:



- **Performance:** Response time, throughput, or latency targets
- **Security:** Protection against unauthorized access and security threats
- **Usability:** Ease of use and user experience
- **Reliability:** System stability, availability, and failure recovery
- **Maintainability:** Ease of updating, modifying, and fixing the system
- **Portability:** Ability to run across different environments or platforms
- **Scalability:** Ability to handle increasing users, traffic, or data

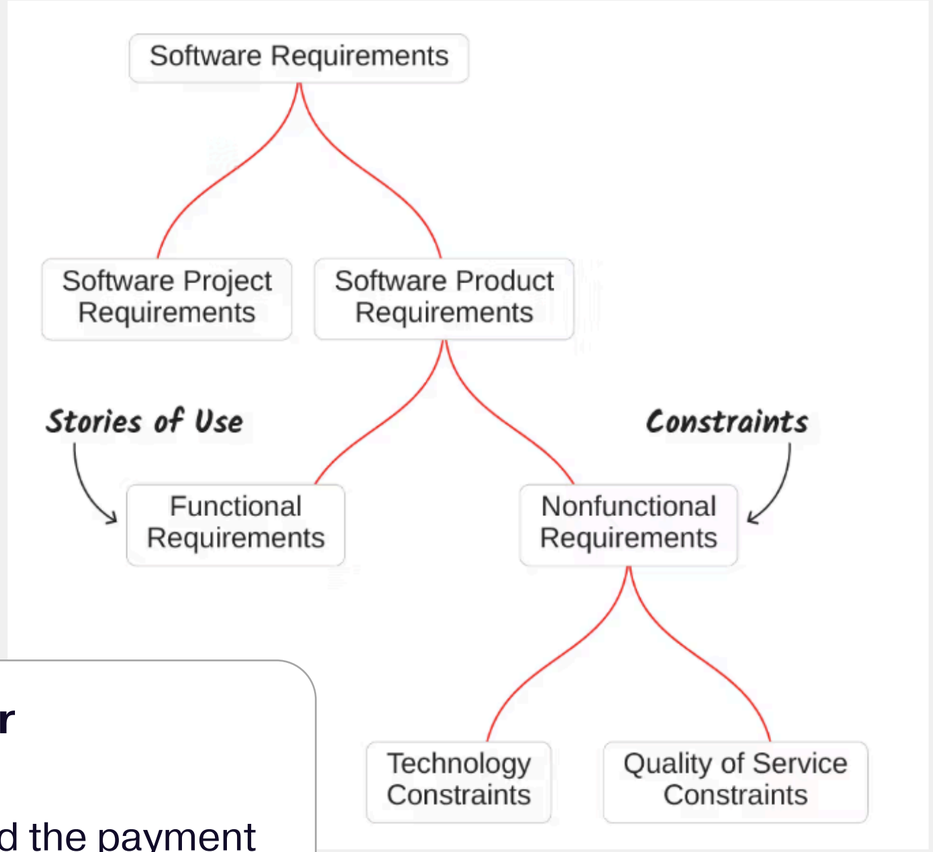
Quality requirements: Car Rental examples

Performance	**95% of searches finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.**
Availability	**The reservation service is available for at least 99.5% of each calendar month, including planned downtime.**
Usability	**At least 4 of 5 first-time test participants complete the standard reservation flow within 3 minutes without help.**

All numerical targets are illustrative assumptions.

Constraints

Limitations of a system, such as speed, scalability, security, and usability.
Often fall under NFR / quality requirement category.



Imposed limit

“The existing payment provider must remain.”

This limits the available solution choices.

Functional behaviour

“The system shall record the payment outcome for a reservation.”

This states what the system must do.

A constraint must come from the brief or an explicitly labelled assumption.

Requirements elicitation - where do we search for business needs?

Requirements elicitation - can you guess?

Ask stakeholders about decisions, exceptions and needs.

Watch how staff handle real work and workarounds.

Inspect forms, policies and existing records.

Compare needs and resolve disagreements together.

For QuickBite, use the company brief and your saved M01 decisions.

Requirements elicitation

Interview

Ask stakeholders about decisions, exceptions and needs.

Observation

Watch how staff handle real work and workarounds.

Document review

Inspect forms, policies and existing records.

Workshop

Compare needs and resolve disagreements together.

For QuickBite, use the company brief and your saved M01 decisions.

Elicitation: a car rental example

Branch employee: “We sometimes confirm a car that is already reserved.”

Clarifying questions:
When does a reservation block availability?
What happens when two requests arrive together?
Do cancelled reservations release the car?

Candidate requirement: prevent overlapping confirmed reservations for the same vehicle.

What kind of requirement is candidate’s requirement?

Elicitation: a car rental example

Branch employee: “We sometimes confirm a car that is already reserved.”

Clarifying questions:
When does a reservation block availability?
What happens when two requests arrive together?
Do cancelled reservations release the car?

Candidate requirement: prevent overlapping confirmed reservations for the same vehicle.

What kind of requirement is candidate’s requirement? **- Functional requirement**

Why?

FR - Functional Requirement:
The system shall prevent overlapping confirmed reservations for the same vehicle.

A related NFR would be something like:
The system shall check reservation conflicts within 1 second.

Requirement quality

Clear

Use precise terms and define important statuses.

Focused

State one expectation at a time.

Consistent

Respect the agreed scope and other requirements.

Verifiable

Describe an observable result or measurable target.

Unclear and clearer wording

Unclear

The system should handle bookings quickly and easily.

Several expectations and no clear measure.

Clearer

The system shall show a reservation reference after successful reservation creation.

One observable outcome.

Acceptance criteria

An acceptance criterion states **observable evidence** that **a requirement has been met.**

Requirement: exclude vehicles under maintenance from search results.

Criterion: with one available vehicle and one vehicle marked under maintenance, a search for the same branch and period returns only the available vehicle.

Success, failure and boundary cases

Success
when we success :)

A vehicle has no overlapping confirmed reservation. The system accepts an otherwise valid reservation.

Failure
...when we fail :)

A vehicle already has a confirmed reservation for an overlapping period. The system rejects the new reservation.

Boundary
Tests behavior exactly at the edge of a rule.

A reservation ends exactly when another starts. Under the assumed back-to-back rental policy, the system accepts the new reservation.

Priorities

Must

Essential for the agreed initial scope.

Prevent confirmed double booking.

Should

Important but can wait if time is limited.

Send a cancellation notification.

Could

Useful if the team has capacity.

Let customers save a preferred branch.

Sources and stable IDs

Source

Brief statement
Saved M01 stakeholder or scope decision
Explicitly labelled working assumption

ID

FR-01 for a functional requirement
NFR-01 for a quality requirement
Keep the ID when wording changes

Requirements register: functional example

Car rental example

ID and type	FR-02 / Functional
Requirement	The system shall reject a reservation that overlaps an existing confirmed reservation for the same vehicle.
Source	Car rental brief: double booking can happen.
Priority	Must
Acceptance criterion	With a confirmed rental from 10 to 12 May, a new request for the same vehicle from 11 to 13 May is rejected.

The register records expectations and planned checks. It does not prove implementation.

Requirements register: quality example

Car rental example

ID and type	NFR-01 / Performance
Requirement	At least 95% of searches shall finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.
Source	Customer need for prompt availability.
Priority	Should
Acceptance criterion	In a 10-minute load test with those conditions, at least 95% of valid search requests return results within 2 seconds.

The numerical targets and test conditions are illustrative teaching assumptions.

QuickBite: Session 3 requirements

M02 Task 2 is today’s required work.

Write 6 functional requirements and 3 non-functional requirements for your own scope.

Link each requirement to at least one relevant business objective from M01.

For each requirement record an ID, source, priority and testable acceptance criterion.

Keep each statement clear and focused on one expectation.

Optional M02 tasks are extra practice. Tasks 6 and 7 belong to Session 4.

Saving and the next checkpoint

Write the register on the M02 Task 2 card and click Save answer.

Your requirements register is the starting point for later models and tests.

Before leaving, export the complete project ZIP and keep it outside the classroom computer.

Session 4 continues M02 with a user scenario and links between objectives and requirements.

Checkpoint 1 covers M01 and M02. No need to upload assesment to Moodle - there’s still one Session ahead (Session 4) which ends our Checkpoint 1.

THANK YOU